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FAO crop coefficient and plant-available water: Do soil and plant hydraulic properties matter?

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Abstract

How to ensure that "the availability of soil water does not limit plant growth or transpiration?" Determining the so-called crop coefficient (K_c) relies on this assumption of optimal or non-limiting soil water conditions, which, in turn, is understood as the soil water content at field capacity, a parameter with, de facto, no relation to plants or transpiration. Many studies on crop evapotranspiration ignore the dynamic nature of plant available water (PAW) driven by the soil hydraulic properties (water retention and hydraulic conductivity) and the characteristics of the plant root system. This raises important questions: How can we guarantee that a K_c value is transferable if soil and plant hydraulics are not explicitly considered? And how can we reconcile the practical determination of crop evapotranspiration with the theoretical one using complex water transfer models? While addressing these complexities remains a challenge, recent advances in process-based modeling offer new opportunities to represent soil–plant–atmosphere interactions more accurately, and to refine the criteria underlying K_c and PAW. Integrating such physically based understanding with established approaches may help bridge the gap between theoretical knowledge and practical applications, ultimately supporting more reliable and adaptable methods for estimating crop evapotranspiration.

Plain Language Summary

Farmers and scientists often use a value called the "crop coefficient" to estimate how much water crops need. This value assumes that plants always have enough water in the soil, but that assumption is based on a soil profile property called "field capacity," which does not consider how plants take up water. Our study claims that this common approach may not be accurate because it ignores important soil and plant characteristics. We argue that new models that simulate water movement through the soil and plants can give a better picture of when crops are truly well-watered.

Abbreviations: FAO, Food and Agriculture Organization; FAO56, FAO Irrigation and Drainage paper 56; FC, Field capacity; PAW, plant available water; RAW, readily available water; RWU, root water uptake; SPAC, soil–plant–atmosphere continuum; TAW, total available water.

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These models consider how roots and soil interact and can improve how we estimate crop water needs. This is important for using water more efficiently in agriculture, especially in a changing climate.

1 | INTRODUCTION

The crop coefficient (K_c), defined as the ratio between crop evapotranspiration (ET_c) and reference evapotranspiration (ET_o), is the foundation of irrigation scheduling and crop modeling worldwide (Allen et al., 1998). However, the K_c concept relies on the assumption of "well-watered" conditions, often equated to maintaining soil water content near field capacity (FC), a soil profile property defined based on drainage properties and without relation to plant transpiration. This apparent oversimplification neglects the dynamic interactions within the soil–plant–atmosphere continuum (SPAC), particularly the role of soil and plant hydraulic properties in regulating water availability and root water uptake (RWU).

Although standard K_c is intended to represent potential crop evapotranspiration under non-limiting conditions, many experimental studies that claim to determine the standard K_c, in fact, estimate the actual crop coefficient (K_c act). The latter reflects the combined influence of management practices, soil constraints, and environmental stress. Empirical evidence suggests that K_c values can be sensitive to environmental factors such as evaporative demand, wind speed, and soil water availability, even under supposedly optimal conditions (Gonçalves et al., 2024; Marin et al., 2020). These findings challenge the assumption that coefficients derived from specific experimental conditions are broadly transferable across different climates and soil types, highlighting the limitations of relying on static thresholds to define plant-available water (PAW).

Meanwhile, advances in soil–plant hydrology have enabled the use of process-based models that more accurately simulate water movement in the SPAC by integrating soil hydraulic properties and plant hydraulics (Couvreur et al., 2012; de Jong van Lier et al., 2013; Vanderborght et al., 2023, 2024). These models provide a more mechanistic understanding of RWU, surpassing the limitations of empirical stress functions and enabling improved estimation of crop transpiration under varying conditions.

In this study, we critically examine the assumptions and practical applications of K_c and PAW by considering current knowledge in soil and plant hydraulics. We argue that the prevailing reliance on empirical thresholds is conceptually inconsistent and often physically unjustified. We discuss the potential of process-based approaches to clarify the definition of "well-watered" conditions and improve the estimation of crop transpiration across diverse environments.

2 | MODELING CROP EVAPOTRANSPIRATION: THE FAO METHOD

2.1 | Crop evapotranspiration under standard conditions

The FAO Irrigation and Drainage paper 56 (FAO56; Allen et al., 1998) describes a method for computing evapotranspiration, which was very well accepted worldwide (Pereira, Paredes, Hunsaker, López-Urrea, & Jovanovic, 2021). The FAO56 was a follow-on to the FAO Irrigation and Drainage paper 24 (Doorenbos & Pruitt, 1977), which introduced the two-step crop coefficient–reference evapotranspiration (K_c–ET_o) procedure to estimate crop water requirement. In this approach, ET_o represents the primary weather-induced effects on water consumption described by the Penman–Monteith equation (Monteith, 1965), and the dimensionless K_c scales the ET_o to account for crop-specific influences during the growing season (Pereira et al., 2015).

The K_c represents the combined effects of resistance to diffusion of water vapor from the evaporating surfaces through the laminar boundary layer, resistance to turbulent transfer to the free atmosphere, and relative amount of radiant energy available as compared to the reference crop (Jensen, 2010). For transferability purposes, FAO56 states the concepts of optimal soil water availability and pristine crop conditions as the basis for tabularized K_c values.

The FAO56 also introduced the K_c curve, that is, K_c as a function of time for initial, development, mid-season, and end-season crop growth stages. The standard values for initial K_c (K_c ini) and end-of-season K_c (K_c end) should be adjusted to reflect local conditions. The former depends on the frequency and amount of wetting and the local climate that conditions the evaporation of water from the soil surface, and the latter depends on the climate and the management practices (Pereira et al., 2015). Furthermore, mid-season K_c values require further adjustment when the meteorological conditions deviate strongly from the standard conditions, particularly wind speed and air humidity (equation 62 of FAO56). If the mid-season K_c is not properly

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adjusted, then none of the K_c values can be considered truly transferable.

In addition, the FAO56 presented the dual crop coefficient, that is, $K_c = K_{cb} + K_e$, where K_{cb} is the basal crop coefficient representing primarily plant transpiration, and K_e is the soil water evaporation coefficient representing evaporation from the surface. The tabulated K_{cb} values refer to conditions when "the soil surface is visually dry, so that evaporation from soil is minimal, but where the availability of soil water does not limit plant growth or transpiration" (Allen et al., 2021). The K_e value, however, varies with the fraction of ground cover, the soil wetting and drying history, the amount of water in the evaporation layer of the soil, as well as with the soil hydrodynamics defined in a daily time step (Allen, 2011; Allen et al., 1998, 2005; Pereira, Paredes, Hunsaker, López-Urrea, & Mohammadi Shad, 2021).

Although the separation of crop transpiration (quantified using K_{cb}) and soil evaporation (quantified using K_e) has added some more mechanism to the K_c approach (Allen, 2011), the definition of optimal soil water and pristine crop conditions remains poorly addressed. In the last 20 years, many studies have been developed to determine the actual K_c values, but they are frequently published with insufficient information on the methods and instrumentation, the climatic and soil conditions, or the crops and cultivation practices used (Pereira, Paredes, Hunsaker, López-Urrea, & Mohammadi Shad, 2021).

In this context, the lack of an appropriate estimation of soil water fluxes is one of the main challenges in achieving the transferability of the actual K_c values. This problem arises from the inadequate definition of the soil hydraulic properties and the rooting depth; the incorrect or inaccurate measurements of soil water content, irrigation applications, and rainfall; the poor estimation of deep percolation or capillary rise fluxes; and/or the use of oversimplified or uncalibrated models (Allen et al., 2011; Evett et al., 2012; Pereira, Paredes, Hunsaker, López-Urrea, & Mohammadi Shad, 2021). Adequate computations of soil water fluxes are essential to derive reliable water balance predictions, which support the calculation of both K_e and K_{cb} fractions.

2.2 | Crop evapotranspiration under non-standard conditions

Below a certain level of soil water storage in the root zone, transpiration becomes source (soil) limited and drought stress occurs, affecting the estimation of K_c act. According to FAO56, the K_c under non-standard, stressed water condition is adjusted using the water stress coefficient (K_s), which describes the effect of drought stress on crop transpiration (Allen et al., 1998). Note that K_s impacts only crop transpiration, and adjusting K_{cb} instead of K_c is more appropriate.

Core Ideas

- Field capacity does not reliably represent water availability for crop transpiration.
- Empirical crop coefficients overlook key soil and plant hydraulic properties.
- Process-based models better simulate soil–water-plant dynamics under field conditions.
- Defining "well-watered" should consider soil water fluxes, not static thresholds.
- Improving crop water estimates requires integrating plant physiology with soil hydrology.

Nevertheless, Allen et al. (1998) advocate that in situations where soil evaporation is not a large component of ET_c, adjusting the "single" K_c using K_s provides reasonable results.

The calculation of K_s requires a daily soil water balance (Allen et al., 2005, 2020; Cholpankulov et al., 2008) and the specification of a shape function for K_s versus soil water content (θ) or soil water potential. To address this, the FAO56 presents a piecewise linear function of K_s versus θ, delimiting two fractions of PAW: the total available water, TAW (between the soil water content at field capacity, θ_{FC} and at the wilting point, θ_{WP}), and the readily available water, RAW (between θ_{FC} and the soil water content at the threshold value or limiting point, θ_{LP}). Between θ_{FC} and θ_{LP}, K_s is constant and equal to one, that is, the value of K_c is unaffected. Between θ_{LP} and θ_{WP}, the value of K_s linearly decreases from one ("potential" evapotranspiration) to zero (no transpiration due to permanent wilting; Figure 1).

A similar approach is adopted in the Feddes transpiration reduction function (Feddes et al., 1978), which also defines reduction thresholds but expresses the stress response as a function of soil water pressure head rather than volumetric water content. In this formulation, the critical pressure head values delineate the optimal and limiting ranges for root water uptake, providing a macroscopic representation of the soil–plant interactions under varying soil water conditions. Despite its empirical formulation, the Feddes function has been widely adopted in process-based modeling due to its simplicity and practical usefulness in describing the effects of soil water stress on root water uptake (de Melo & de Jong van Lier, 2021).

In the FAO56 approach, the threshold water content θ_{LP} represents the onset of transpiration reduction (θ_t in Figure 1). Following Allen et al. (1998), θ_{LP} is derived from the average fraction of TAW that can be depleted from the root zone before drought stress occurs (p). Values for p are listed as a function of the crop type and maximum rooting depth in Table 22 of the FAO56. The authors highlight that "a value of 0.50 for p is

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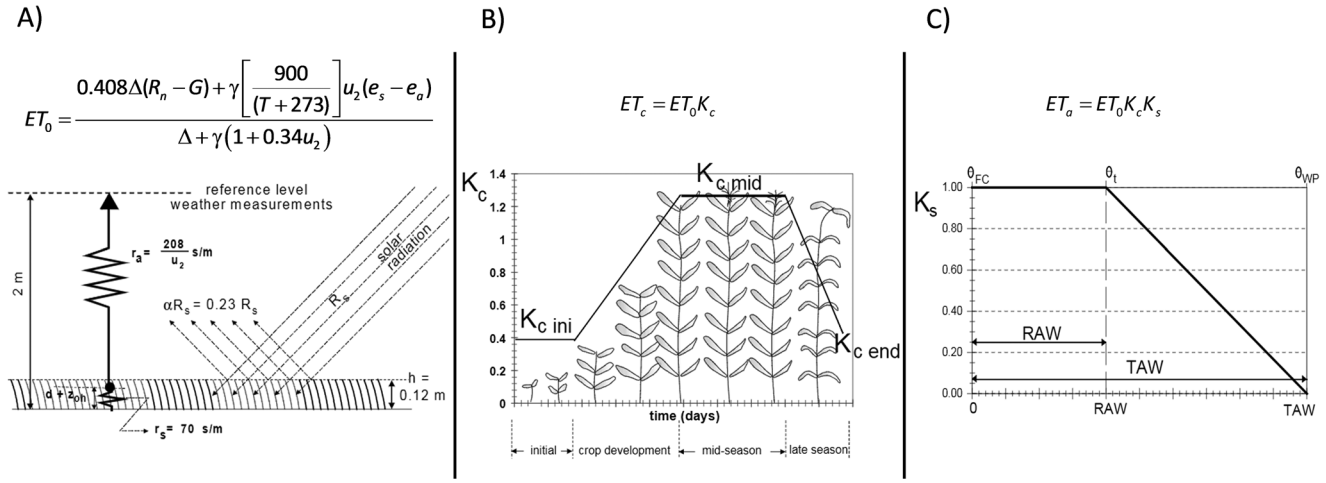


FIGURE 1 A summary of the FAO method to compute crop evapotranspiration. (A) The Penman–Monteith equation to estimate evapotranspiration of a reference crop (ET_o), derived from the heat balance and sensible and latent heat flux equations. (B) The estimation of crop evapotranspiration (ET_c) under well-watered conditions using the crop coefficient (K_c) as a function of time (the K_c curve). (C) The estimation of actual crop evapotranspiration (ET_a) under suboptimal conditions by adjusting K_c using the water stress coefficient (K_s) as a function of the soil water content. Equations and figures taken from the FAO Irrigation and Drainage paper 56 (FAO56) bulletin (Allen et al., 1998).

commonly used for many crops." Besides, p is also a function of the evaporative demand of the atmosphere, and the authors suggested empirical adjustments for low and high rates of ET_c, taking a rate of about 5 mm day^{-1} as the basis for the tabulated p values.

Allen et al. (1998) recognized that "to express the tolerance of crops to water stress as a function of the fraction (p) of TAW is not wholly correct." They agree that the rate of root water uptake is more directly influenced by the soil hydraulic properties than by the soil water content; thus, the value of p is also a function of the soil type. Nonetheless, most users of the FAO56 approach treat p simply as a crop parameter, implicitly assuming that soil hydraulic properties do not matter. This misapplication can result in an oversimplified representation of water stress effects, particularly when estimating actual evapotranspiration.

Many applications of the FAO56 also fail to clearly distinguish between its two conceptual components: the potential component, represented by the standard crop coefficient (K_c), and the stress-adjustment component, represented by the water stress coefficient (K_s). Together, these determine the actual crop coefficient (K_c act = $K_c \bullet K_s$). Misinterpretations often occur when K_c act is estimated empirically without explicitly accounting for soil hydraulic properties. This distinction is particularly important because the FAO56 is often used as a comparative reference for quality control applications (Pereira et al., 2015) and serves as a boundary condition in many process-based models, integrating empirical to physically based representations of crop water use. Despite its simplifications, it remains the most used method to compute crop water and irrigation requirements.

3 | AFTER ALL, WHAT IS A "WELL-WATERED" CONDITION?

In an update to the FAO56 regarding the standard single and basal crop coefficients for field crops, Pereira, Paredes, Hunsaker, López-Urrea, and Mohammadi Shad (2021) emphasize that "ET_c and K_c refer to optimal, well-watered conditions and pristine cultivated crops." The expression "well-watered" is, in fact, a recurring term in studies focused on ET_c estimation (Abioye et al., 2020; Alexandris & Proutsos, 2020; Allen et al., 2011, 2021; Colmer & Barton, 2017; Cuxart & Boone, 2020; Pereira et al., 2015, 2020; Pereira, Paredes, Hunsaker, López-Urrea, & Mohammadi Shad, 2021; K. Wang & Dickinson, 2012; T. Wang et al., 2021, 2025). In practice, this ideal water condition is usually defined by assuming that there is no drought stress for plants if the soil water content is kept at or near the FC (Christopher et al., 2016; del Pozo et al., 2020; N. Kumar et al., 2020, 2021; Medici et al., 2014). If FC is associated with a fixed pressure head, as is common practice, this assumption becomes conceptually similar to the Feddes root water uptake reduction function. These assumptions, however, have no sound physical basis, since FC is a soil profile property originally determined as a function of the drainage of non-vegetated soils (Logsdon, 2019), hence, without the presence of transpiration.

Although the very concept of FC as the upper limit of PAW is subject to criticism (de Jong van Lier, 2017; Logsdon, 2019), the methods for defining FC and, consequently, the "well-watered" condition, may be a major concern. A common practice worldwide is to adopt a value of pressure head usually between −1 m (pF 2.0) and −3.3 m (pF 2.5 or one-

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third bar) to correspond to FC (de Jong van Lier & Wendroth, 2016; Logsdon, 2019; Nemes et al., 2011; Turek et al., 2023). Although the pressure head corresponding to FC should, in principle, depend on soil texture and profile properties, it is often applied in a simplified manner without explicit consideration of soil-specific hydraulic properties. These static values of FC are an oversimplification that may lead to discrepancies and uncertainties due to the dynamic nature of the flow processes involved (Inforsato & de Jong van Lier, 2021; Liang et al., 2016).

Some approaches have been proposed to retrieve the original concept of FC based on water fluxes in the soil (Assouline & Or, 2014; Inforsato & de Jong van Lier, 2021; Meyer & Gee, 1999; Nasta et al., 2023; Zacharias & Bohne, 2008). Other studies, however, have aimed to optimize static FC criteria using dynamic hydrological models as benchmarks (de Jong van Lier & Wendroth, 2016; H. Kumar et al., 2023; Lena et al., 2022; Romano et al., 2011; Turek et al., 2022). We consider that both research lines may improve the estimation of PAW and crop evapotranspiration, but concepts of FC based on soil water fluxes have the advantage of a straightforward flux-based definition of FC (de Melo et al., 2023; Inforsato & de Jong van Lier, 2021).

The assessment of flux-based FC requires the determination of site-specific soil hydraulic properties comprising the soil water retention function relating pressure head and θ, and the hydraulic conductivity function relating pressure head or θ with hydraulic conductivity. As such determination can be expensive, time-consuming, and not readily available for many soils, this is a common justification for using empirical models instead of process-based models to assess water transfer processes (Jarvis et al., 2022). However, this practice seems unjustified as many practical measurement technologies are currently available to determine both the retention and hydraulic conductivity functions (Peters & Durner, 2008; Vereecken et al., 2015). Alternatively, modeling techniques, for example, inverse modeling, and integration of machine learning with process-based models, can be used when direct measurements are impractical, recognizing that these approaches introduce additional uncertainty. Although technically demanding, such methods are indispensable for high-impact, reliable, and detailed studies aiming to accurately represent soil–plant–atmosphere interactions.

Encouraging process-based modeling to assess crop evapotranspiration, mainly to elucidate transpiration mechanisms based on water fluxes in the soil–plant system, seems promising to clarify what is a "well-watered" condition instead of assuming this condition based on arbitrary criteria. We are aware that this definition will not be achieved by performing just a few studies to identify transpiration restriction mechanisms that can be very complex (Burridge et al., 2022). Nevertheless, at least we can get insights in this direction and contribute to overcoming "the elephant in the room"—the empirical models (Jarvis et al., 2022)—in soil–crop modeling research.

4 | IMPROVED REPRESENTATIONS OF THE SOIL–PLANT SYSTEM

Understanding crop water use requires a mechanistic approach that captures the complex interactions between soil properties, plant physiology, and environmental drivers. While traditional models can be useful under standard conditions, they fall short in representing the dynamic and spatially variable processes that govern water uptake under field conditions. Process-based models of root water uptake incorporate the physical principles of soil water flow and plant hydraulics, allowing for a more detailed and realistic simulation of how water moves from the soil through the plant to the atmosphere. These models account for the heterogeneity in soil moisture, root distribution, and plant water status, making them valuable for evaluating crop responses under water-limited environments (de Melo et al., 2025). However, such models should not be seen as practical, direct substitutes for FAO56, but rather as complementary research tools that can inform the development and calibration of process-based simplified approaches.

Process-based RWU models to simulate crop transpiration mechanisms are not new in the literature. One of the first attempts at modeling RWU is attributed to Gardner (1960), who solved soil water flow equations for a single root extracting water from the soil. The author verified the effects of the soil water diffusivities, potentials and contents, and the root radius on the rate of crop transpiration. A few years later, Cowan (1965) further developed Gardner's theory by suggesting a new model for the plant hydraulic behavior considering the soil depth, the rooting density, the internal plant resistance to water flow, and a critical value for the leaf water potential representing stomata closure.

Since then, several process-based RWU models have been developed (Bouda & Saiers, 2017; Couvreur et al., 2012; Couvreur, Vanderborght, Draye, et al., 2014; de Jong van Lier et al., 2008; de Jong van Lier et al., 2013; de Willigen & van Noordwijk, 1987; Doussan et al., 2006; Hillel et al., 1975; Javaux et al., 2008, 2013; Mariño & Tracy, 1988; Moldrup et al., 1992; Nimah & Hanks, 1973; Vanderborght et al., 2021, 2023, 2024; Wu et al., 1999). Although there are differences in the description of water flow and root system properties, ranging from a single flow path in a lumped root zone to detailed representations of the 3D root system architecture, all approaches include soil and plant hydraulic parameters in their modeling approaches.

Reviews of process-based approaches to simulating RWU classify the models into microscopic and macroscopic scales (R. Kumar et al., 2015; Molz, 1981; Raats, 2007). At the

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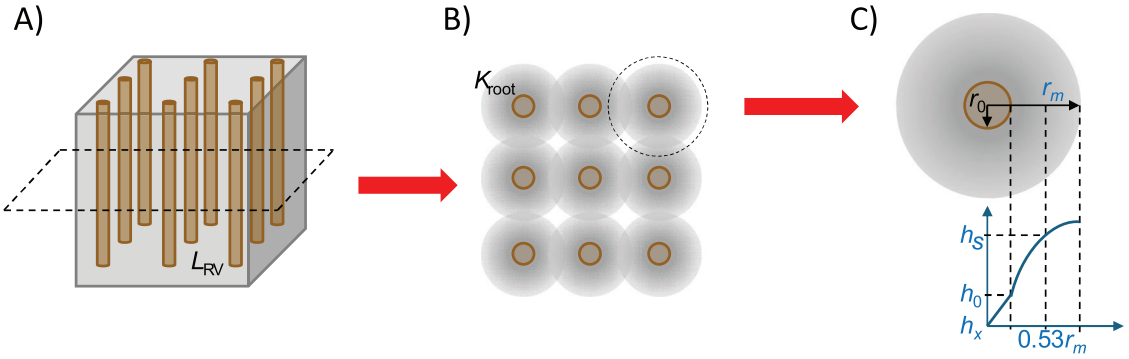


FIGURE 2 An upscaled microscopic root water uptake model by simplification of root system architecture. (A) Root system as equally spaced cylinders based on root length density (L_{RV}) of a specific soil layer. (B) Rhizosphere radius, where water moves radially toward the root center at a rate determined by the root tissue hydraulic conductivity (K_{root}). (C) Radial flow in the rhizosphere (r_m), with mean soil pressure head h_s corresponding to water potential at $0.53r_m$. r_0, h_0, and h_x are root radius, water potential at the root surface, and water potential in the root xylem, respectively. Schemes based on Vanderborght et al. (2023).

first scale, RWU is represented by a diffusion equation, which describes the water flow from the soil to a single root from suitable initial and boundary conditions. At the second scale, RWU is represented by a sink term in the volumetric mass balance, usually described by the Richards equation (Raats, 2007). In many cases, upscaling microscopic models is required to apply them to real-world problems, including field crops. In this sense, the information provided by these single-root models enables the formulation of simple, process-based macroscopic models with practical applications (Couvreur et al., 2012; de Jong van Lier et al., 2008, 2013; Vanderborght et al., 2024).

The available process-based RWU models coupled with crop growth models at the field scale (de Melo et al., 2025; Heinen et al., 2024) focus on the soil hydraulics but simplify the root system architecture (Figure 2). However, the contribution of root architecture to RWU is hardly separable from the root and soil hydraulic properties (Draye et al., 2010). On the other hand, models with a detailed description of the root architecture (Doussan et al., 2006; Javaux et al., 2008; Leitner et al., 2010) are challenging to use at the field scale. The main reason is the need for explicit 3D root architecture information, where each root segment can differ in plant physiological properties, for example. radial and axial hydraulic conductivities and root age (Schnepf et al., 2018). Furthermore, the simulation of 3D water flow requires very high spatial discretization to resolve the hydraulic gradients near the soil–root interface, which is computationally expensive (Khare et al., 2022). Therefore, 3D root hydraulic models should be primarily viewed as research tools that enhance process-based understanding and provide theoretical basis for simplifying and parameterizing 1D or empirical models.

Approaches to upscale 3D root hydraulic models to 1D RWU models have been developed by combining the soil and root hydraulics (Couvreur, Vanderborght, Beff, et al., 2014; Vanderborght et al., 2021, 2023, 2024) and implementing them in models that simulate 1D soil water flow (Cai et al., 2018; Sulis et al., 2019) and crop growth (Nguyen et al., 2020, 2022). Nevertheless, all these process-based RWU models, coupled or not with agro-hydrological models, are still not widely used compared to empirical models, even if dismissing parameter fitting (Vanderborght et al., 2023). This limited use reflects a broader paradox in modeling practice: while process-based models are conceptually more robust and better grounded in biophysical principles, their widespread application is hindered by the scarcity of accurate RWU measurements needed for validation. Bridging this gap will depend on integrating mechanistic understanding from such advanced models into simpler, scalable frameworks that can enhance the predictive capacity and reliability of methods such as the FAO56 under diverse field conditions.

5 | CONCLUSION

The FAO56 approach is the preferred method for practical computation of crop evapotranspiration under standard ("well-watered") and non-standard ("suboptimal") conditions. Nonetheless, its empirical formulation for computation of PAW does not explicitly incorporate soil and plant hydraulic properties, which are key to understanding water transfer in the soil–plant system. Advances in process-based root water uptake models, coupled with field-scale agro-hydrological models, now offer opportunities to refine these criteria, addressing mechanisms beyond the scope of empirical approaches. We encourage the research community to explore the integration of these process-based approaches to complement and strengthen the method of the FAO56 for computing crop water and irrigation requirements.

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AUTHOR CONTRIBUTIONS

Marina Luciana Abreu de Melo: Conceptualization; formal analysis; investigation; methodology; writing—original draft; writing—review and editing. **Quirijn de Jong van Lier**: Conceptualization; project administration; supervision; validation; visualization; writing—review and editing. **Fábio Ricardo Marin**: Validation; visualization; writing—review and editing. **Jos C. van Dam**: Supervision; validation; visualization; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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